

**5. Department of Science and Technology – VI
Regional Standards and Testing Laboratory**

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Field of Testing : Chemical Testing
 Microbiological Testing

Chemical Testing

I. Foods		
.01 Cereals and Cereal Products		
1. Breakfast cereals	Crude Protein	Kjeldahl Block Digestion and Distillation Method (AOAC 2001.11, 21st ed.)
2. Cereal/cereal grains	Crude Fat	Soxhlet Method using petroleum ether with acid hydrolysis (In-house Method)
3. Cultured seeds and grains	Total Fat	
4. Soya flours concentrates and isolates	Moisture	Oven Method (AOAC 925.09, 925.10, 930.15 21st ed.)
5. Flour, corn meal, corn grits, semolina	Ash	Gravimetry (AOAC 923.03 21st ed.)
6. Frozen entrees containing rice or corn flour	pH	Electrometric method (AOAC 943.02, 21st ed.)
7. Soy protein	Sodium	Dry Ashing and Microwave Plasma AES (In-house method)
8. Tofu	Carbohydrates and Energy	Calculation, (AOAC 986.25E, 21st ed.)
9. Pasta products and noodles (e.g. rice paper, rice vermicelli, soybean pastas and noodles)	Nutrition facts Evaluation	Calculation, USFDA and 2015 PDRI (FDA Circular No. 2023-009)
10. Starch		
.03 Dairy Products		

1. All cheese made from pasteurized milk (cottage cheese, soft & semi-solid cheese)	Crude Protein	Kjeldahl Block Digestion and Distillation Method (AOAC 991.20, 930.29, 21st ed.)
2. Processed cheese spread		
3. Ice cream and sherbet plain and flavoured		
4. Ice cream with added ingredients		
5. Flavored ice		
6. Milk powders (whole, nonfat or filled milk, buttermilk, whey & whey protein concentrate)		
7. Sweetened Condensed milk		
8. Liquid Milk (evaporated or Ready To Drink) and Cream (Ultra Heat Temperature/sterilized)	Carbohydrates and Energy	Calculation, (AOAC 986.25E, 21st ed.)
9. Pasteurized milk		
10. Pasteurized cream	Nutrition Facts Evaluation	Calculation, USFDA and 2015 PDRI (FDA Circular No. 2023-009)
11. Yogurt and other fermented milk		
.04 Meat and Meat Products		
1. Dried animal products	Crude Protein	Kjeldahl Block Digestion and Distillation Method (AOAC 991.20, 21st ed.)
2. Meat paste and pate' (heat treated)		
3. Cold cuts, frozen and chilled hotdogs, corned beef, luncheon meat		
4. Packaged cooked cured/salted meat (ham, bacon)		
5. Fermented, comminuted meat, not cooked (dry and semi-dry fermented sausages)		
6. Cooked poultry meat, frozen to be re-heated before eating (e.g. prepared frozen meals)		
	Crude Fat	Soxhlet Method using petroleum ether with acid hydrolysis (In-house Method)
	Total Fat	
	Moisture	
	Ash	Oven Method (AOAC 950.46 21st ed.)
		Gravimetry (AOAC 920.153, 21st ed.)

7. Cooked poultry meat, frozen, ready-to-eat (e.g. turkey rolls)	Sodium	Dry Ashing and Microwave Plasma AES (In-house method)
8. Cured/smoked poultry meat	Carbohydrates and Energy	Calculation, (AOAC 986.25E, 21st ed.)
9. Dehydrated poultry products		
10. Fresh/frozen raw chicken (during processing)	Nutrition Facts Evaluation	Calculation, USFDA and 2015 PDRI (FDA Circular No. 2023-009)
11. Meat products in hermetically sealed containers		
.06 Sugar and Sugar Products		
1. Refined and raw sugars	Crude Protein	Kjeldahl Block Digestion and Distillation Method (AOAC 991.20, 21st ed.)
2. Brown sugar	Crude Fat Total Fat	Soxhlet Method using petroleum ether with acid hydrolysis (In-house Method)
3. Sugar solutions and syrups	Moisture	Oven Method(AOAC 925.45, 21st ed.)
4. Other sugars and syrups (e.g. xylose, maple syrup, sugar toppings)	Ash Sodium	Gravimetry (AOAC 900.02, 21st ed.) Dry Ashing and Microwave Plasma AES (In-house method)
5. Honey	Carbohydrates and Energy	Calculation, (AOAC 986.25E, 21st ed.)
6. Table-top sweeteners, including those containing high-intensity sweeteners	Nutrition facts Evaluation	Calculation, USFDA and 2015 PDRI (FDA Circular No. 2023-009)
.07 Confectioneries		
1. Cocoa powder	Crude Protein	Kjeldahl Block Digestion and Distillation Method (AOAC 991.20, 21st ed.)
2. Chocolate products	Crude Fat Total Fat	Soxhlet Method using petroleum ether with acid hydrolysis (In-house Method)
3. Chocolate confectioneries (chocolate bars, blocks, bonbons)	Moisture	Oven Method (AOAC 925.45, 21st ed.)

	Ash	Gravimetry (AOAC 900.02, 21st ed.)
	Sodium	Dry Ashing and Microwave Plasma AES (In-house method)
4. Sugar confectioneries (hard and soft candies, toffees, caramel, fondants, creams, nougats and pastes)	Carbohydrates and Energy	Calculation, (AOAC 986.25E, 21st ed.)
	Nutrition facts Evaluation	Calculation, USFDA and 2015 PDRI (FDA Circular No. 2023-009)
.12 Fruit Juices, drinks and concentrates		
3. Water-based flavoured drinks (Carbonated, Non-carbonated, Concentrates (liquid or solid))	pH	Electrometric method (AOAC 943.02, 21st Ed)
	Total Sugars	Lane-Eynon method (AOAC 968.28, 21st Ed.)
	Total Titratable Acidity	Titrimetric method, (AOAC 942.15, 21st Ed.)
VI. Water		
.01 Bottled water	Calcium	3000 Ca SMEWW 23rd Ed., 2017
	Chloride	4500 Cl SMEWW 23rd Ed., 2017
	pH	4500 H SMEWW 23rd Ed., 2017
	Total Dissolved Solids	2540 C SMEWW 23rd Ed., 2017
	Total Hardness	2340 C SMEWW 23rd Ed., 2017
	Conductivity	2510 SMEWW 23rd Ed., 2017
	Mg by calculation	3500 Mg SMEWW 23rd Ed., 2017

Microbiological Testing

I. Foods		
1.0 Milk and Dairy Products		
01.1 Milk Powders	Aerobic Plate Count Yeast and Mold Count Coliform Count E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN BAM/MPN BAM/Direct Plate Count AOAC 2014.01/3M Petrifilm SALX
01.2 Sweetened condensed milk/Creamer		
01.3 Liquid milk (evaporated or ready to drink) and cream (UHT/sterilized)		
01.4 Pasteurized milk		
01.5 Pasteurized cream		
01.6 Yogurt and other fermented milk		
01.7 Heat treated fermented milks (sterilized, UHT)		
01.8 Processed cheese spread		
01.9 Soft Cheese (from pasteurized milk)		
01.10 Hard and semi-hard cheese		
01.11 All raw milk cheese; raw milk unripened cheese with moisture > 50%, pH > 5.0		
01.12 Cream cheese products		
02.0 Fats and oils, and fat emulsions		
02.1 Butter (whipped, pasteurized)	Aerobic Plate Count Yeast and Mold Count Coliform Count E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN BAM/MPN BAM/Direct Plate Count AOAC 2014.01/3M Petrifilm SALX
02.2 Butter made from unpasteurized milk or unpasteurized milk products		
02.3 Margarine		
02.4 Virgin Coconut Oil		
04.0 Confectioneries		
04.1 Cocoa powder	Aerobic Plate Count Yeast and Mold Count	BAM/Pour Plate Method BAM/Spread Plate Method
04.2 Chocolate products		

04.3 Chocolate confectioneries (chocolate bars, blocks and bonbon)	Coliform Count E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/MPN BAM/MPN BAM/Direct Plate Count AOAC 2014.01/3M Petrifilm SALX
04.4 Sugar confectioneries (hard and soft candies, toffees, caramel, fondants, creams, nougats and pastes)		
04.5 Ethnic Milkbased Confectioneries (Pastillas and Yema)	Aerobic Plate Count Yeast and Mold Count Coliform Count E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN BAM/MPN BAM/Direct Plate Count AOAC 2014.01/3M Petrifilm SALX
04.6 Ethnic Flourbased Confectioneries (Polvoron, Piaya and Barquillos)		
04.7 Coconut Sap Sugar		
04.8 Dry, Granulated Sugar		
04.9. Liquid Sugar (sugar syrup)		
04.10 Non-nutritive sweeteners (e.g. aspartame, sucralose) and bulk sweeteners (e.g. polysols)		
08.0 Bakery Products		
08.1 Frozen bakery products (ready to eat) with low acid or high aw fillings or toppings	Aerobic Plate Count Yeast and Mold Count Coliform Count E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN BAM/MPN BAM/Direct Plate Count AOAC 2014.01/3M Petrifilm SALX
08.2 Frozen bakery products (to be cooked) with low acid or high aw fillings or toppings (e.g. meat pies and pizzas)		
08.3 Frozen and refrigerated doughs		

08.4 Baked goods (e.g. breads, cakes, pie crust, pastries and fried doughs)	Aerobic Plate Count Yeast and Mold Count Coliform Count E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN BAM/MPN BAM/Direct Plate Count AOAC 2014.01/3M Petrifilm SALX
08.5 Coated or filled, dried shelf-stable biscuits		
10.0 Meat and meat products		
10.1 Dried animal products	Aerobic Plate Count Yeast and Mold Count Coliform Count E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN BAM/MPN BAM/Direct Plate Count AOAC 2014.01/3M Petrifilm SALX
10.2 Meat paste and pate (heat treated)		
10.3 Cold cuts, frozen and chilled hotdogs, corned beef, luncheon meat		
10.4 Packaged cooked cured/salted meat (ham, bacon)		
10.5 Fermented, comminuted meat, not cooked (dry and semi-dry fermented sausages)	Aerobic Plate Count Yeast and Mold Count Coliform Count E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN BAM/MPN BAM/Direct Plate Count AOAC 2014.01/3M Petrifilm SALX
10.6 Cooked poultry meat, frozen to be re- heated before eating (e.g. prepared frozen meals chicken burgers, chicken turkey rolls, chicken nuggets, other breaded poultry meat products)		
10.7 Cured/smoked poultry meat		
10.8 Dehydrated poultry products		

10.9 Meat products in hermetically sealed containers		
10.10 Marinated Meat Products e.g. Marinated meat and meat preparations (tapa, sisig, etc.), Marinated Poultry, Dim Sum made from meat (siomai)	Aerobic Plate Count Yeast and Mold Count Coliform Count E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN BAM/MPN BAM/Direct Plate Count AOAC 2014.01/3M Petrifilm SALX
10.11 Minced meat and meat preparations made from poultry meat intended to be eaten cooked		
10.12 Minced meat and meat preparations made from species other than poultry intended to be eaten cooked		
10.13 Foods cooked immediately prior to sale or consumption e.g. Takeaway food, burgers, kebabs, sausages, pizza, ready meals (cook/chill and cook/freeze) after regeneration		
11.0 Fish and fish products		
11.1 Fresh frozen fish and cold-smoked	Aerobic Plate Count Yeast and Mold Count Coliform Count E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN BAM/MPN BAM/Direct Plate Count AOAC 2014.01/3M Petrifilm SALX
11.2 Pre-cooked breaded and fish		
11.3 Frozen raw crustaceans		
11.4 Frozen cooked crustaceans		
11.5 Cooked crustaceans	Aerobic Plate Count Yeast and Mold Count Coliform Count	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN

11.6 Cooked, chilled and frozen crabmeat	E. coli Count S. aureus (coagulase positive) Count Detection of Salmonella	BAM/MPN BAM/Direct Plate Count
11.7 Fresh and frozen bivalve molluscs		AOAC 2014.01/3M Petrifilm SALX
11.8 Fish and shellfish products, cooked crustaceans in hermetically sealed containers (thermally processed) e.g. cooked bagoong/shrimp paste		
11.9 Ethnic food products – Dried, salted fish		
11.10 Smoked Fish		
11.11 Salt Fermented Fish and Shrimps (bagoong)		
11.12 Thermally processed fish products		
11.13 Aquatic products – cooked (heat-treated) aquatic product; intended for human consumption aquatic plant products (algae)		
11.14 Fish and crustacean based processed meat e.g. fish ball, squid ball		
13.0 Beverages		
13.1 Non-alcoholic beverages (e.g. Ready to drink softdrinks, iced tea, energy drinks)	Aerobic Plate Count Yeast and Mold Count Coliform Count E. coli Count	BAM/Pour Plate Method BAM/Spread Plate Method BAM/MPN

13.2 Frozen juice concentrate	S. aureus (coagulase positive) Count	BAM/MPN BAM/Direct Plate Count
13.3 Juices in hermetically sealed containers (Tetra Pack etc.)		
13.4 Powdered beverages (e.g. iced tea, powdered juices/mixes)		
13.5 Fruit beverage products		
13.8 Chilled Young Coconut Water (Buko Juice)		
IV. Water		
13.6 Bottled Water (Purified/Distilled Water) 13.7 Bottled Water (Mineral Water)	Heterotrophic Plate Count	SMEWW 23rd Ed./ Pour Plate Method 9215 B
	Total Coliform Count	SMEWW 23rd Ed./ MPN Method 9221 B
	Fecal Coliform Count	SMEWW 23rd Ed./ MPN Method 9221 E
	E. coli Count	SMEWW 23rd Ed./ MPN Method 9221 G

Legend to Reference Methods

AOAC – Association of Official Analytical Chemists

BAM - Bacteriological Analytical Manual

SMEWW - Standard Methods for Examination of Water and Wastewater

***** *Nothing Follows* *****